

BLACK DIAMOND, AB, Canada

WMO: 719820

Lat: 50.7067N

Lon: 114.1517W

Elev: 3793

StdP: 12.79

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-20.7	-15.0	-26.4	1.5	-20.4	-20.7	2.0	-14.7	28.9	31.2	25.4	31.6	4.7	250	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.7	82.8	59.3	79.1	58.6	75.8	57.3	62.9	75.3	61.3	74.1	59.6	72.1	9.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.3	83.6	69.0	56.3	77.7	67.0	54.4	72.6	64.4	30.5	75.8	29.2	74.7	27.9	72.2	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.7	20.0	17.2	-28.6	89.2	5.4	4.8	-32.4	92.7	-35.6	95.5	-38.6	98.2	-42.4	101.6		
(5)			-28.8	66.8	5.3	1.4	-32.6	67.8	-35.7	68.7	-38.7	69.5	-42.5	70.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	19.9	17.1	27.4	37.8	47.8	54.5	59.8	58.8	50.8	39.3	26.7	16.5
	DBStd	18.84	14.42	14.86	13.99	8.63	7.49	5.32	4.78	5.46	8.31	8.92	12.84	14.57
	HDD50	5256	933	922	702	374	132	16	1	4	89	343	701	1039
	HDD65	9824	1398	1342	1166	817	532	316	172	201	429	797	1150	1504
	CDD50	931	0	0	1	6	66	153	303	278	112	11	1	0
	CDD65	23	0	0	0	0	0	2	10	9	2	0	0	0
	CDH74	1143	0	0	0	7	54	92	433	415	142	0	0	0
	CDH80	264	0	0	0	0	3	15	108	112	26	0	0	0
Wind	WSAvg	7.1	7.0	6.3	7.6	8.1	8.0	7.8	7.0	6.2	6.8	7.0	7.2	6.6
Precipitation	PrecAvg	23.5	1.1	1.0	1.4	1.9	3.1	4.2	2.4	2.4	2.0	1.3	1.3	1.1
	PrecMax	36.7	2.0	1.5	3.5	3.6	5.7	14.9	6.3	5.2	5.7	2.4	3.2	2.5
	PrecMin	16.0	0.4	0.2	0.6	1.1	0.8	0.7	0.6	0.4	0.2	0.2	0.1	0.4
	PrecStd	4.3	0.4	0.4	0.7	0.7	1.4	2.7	1.5	1.1	1.3	0.6	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.9	53.3	59.9	73.5	79.8	82.4	87.3	88.4	83.5	70.8	63.5	49.9
		MCWB	39.7	41.9	43.2	49.6	53.9	61.1	60.1	59.1	56.3	50.7	45.6	39.0
	2%	DB	46.6	46.3	55.3	66.1	74.8	76.2	83.0	82.4	77.9	65.0	54.7	44.7
		MCWB	36.8	36.6	41.2	47.0	52.4	57.0	60.8	58.7	56.1	48.3	42.5	36.0
	5%	DB	43.4	41.7	50.9	59.8	70.4	72.5	78.8	78.2	73.6	60.4	48.8	40.8
		MCWB	35.4	33.3	38.9	43.3	50.9	55.4	60.2	58.4	54.5	46.2	38.9	32.9
	10%	DB	39.8	37.4	46.1	54.4	65.8	69.5	74.9	74.1	68.4	55.7	43.9	36.5
		MCWB	33.1	31.1	36.6	40.4	48.7	54.1	59.6	58.2	52.3	43.7	35.7	30.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.8	42.3	45.3	50.2	55.5	63.4	65.9	64.1	59.6	52.4	46.5	39.8
		MCDB	50.8	52.2	56.9	71.9	73.5	75.9	77.6	75.9	76.5	66.9	60.1	48.9
	2%	WB	37.6	36.8	41.9	46.9	53.4	59.7	63.7	62.1	57.5	49.6	43.3	36.2
		MCDB	45.6	45.2	53.6	64.1	72.1	71.4	75.5	73.3	73.2	62.7	54.3	44.4
	5%	WB	35.5	33.9	39.3	44.1	51.6	57.2	62.1	60.5	55.4	46.8	39.3	33.3
		MCDB	42.9	40.8	49.8	58.1	68.3	69.5	74.4	73.6	70.8	59.2	48.1	39.6
	10%	WB	32.9	31.0	36.8	41.2	49.6	55.4	60.5	58.8	53.2	44.0	36.3	30.7
		MCDB	39.7	36.7	45.8	51.6	63.9	66.4	73.0	71.2	66.7	54.8	43.1	36.5
Mean Daily Temperature Range	5% DB	MDBR	20.8	21.7	22.0	23.3	24.7	24.0	26.7	26.6	26.2	24.3	22.1	20.7
		MCDBR	23.9	26.1	28.3	34.5	34.3	30.9	35.8	35.9	36.7	32.4	28.0	23.7
		MCWBR	16.3	17.8	16.8	17.7	15.5	15.1	16.5	15.7	16.9	17.4	16.9	16.5
	5% WB	MCDBR	23.6	25.3	26.7	33.6	31.2	27.6	29.1	28.9	33.2	31.4	27.5	23.9
		MCWBR	16.5	17.9	16.5	17.8	14.6	14.6	16.0	14.8	15.7	17.3	17.1	17.0
Clear-Sky Solar Irradiance	taub	0.251	0.272	0.281	0.337	0.342	0.344	0.340	0.356	0.313	0.278	0.254	0.236	
	taud	2.437	2.419	2.465	2.313	2.372	2.406	2.425	2.384	2.485	2.560	2.511	2.440	
	Ebn at Noon	258	279	297	287	290	289	288	278	281	275	256	247	
	Edh at Noon	18	24	28	37	36	36	34	34	27	21	17	16	
All-Sky Solar Radiation	RadAvg	363	669	1050	1473	1724	1845	2065	1693	1190	736	390	283	
	RadStd	26	50	77	100	137	140	137	118	126	76	30	20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page